Progress Check

Natural Disaster Severity Prediction

Group 5

Hsin-Yu Chen, Wei-Hsin Hung, Sky Shih-Kai Hong

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Problem & Dataset

Goal: predict weekly drought severity for each region over a 5-week horizon after a 91-day blind gap.

Item	Value
Regions	2,248
Training records	12,319,040
Weekly labeled targets	1,746,696
Meteorological features	14
Metric	MAE, lower is better

Key Challenges

- **Sparse labels:** `score` is only observed weekly, so most daily rows have no target.
- **Temporal mismatch:** daily weather signals must explain weekly drought severity.
- **91-day blind gap:** no ground-truth severity score is available before the test horizon.
- **Validation mismatch:** MAE values from different validation strategies are not directly comparable.

Current Implementation

The repository now supports three boosted-tree model families.

1. Build weather, seasonality, rolling, EWMA, climatology anomaly, and drought-proxy features.
2. Use 91-day-gapped historical score features to avoid touching the blind test window.
3. Train five direct horizon models: Week 1 through Week 5.
4. Blend LightGBM, XGBoost, and CatBoost outputs with `src/ensemble.py`.

Engineering status: `src/predict.py` can load LGB/XGB/CatBoost run directories.

Completed Experiments

Experiment	Validation	Local MAE	Public MAE / Status
LGBM v2	Holdout	0.6942	LGB/XGB anchor
XGBoost v1	Holdout	0.7150	LGB/XGB anchor
LGB/XGB 50/50	-	-	0.8232
CatBoost tail2737	Rolling origin	0.2212 / 0.2192 rerun	ensemble member
LGBM micro 20260520	Rolling origin	0.2002	diagnostic only
LGB/XGB/Cat 35/35/30	-	-	**0.8124**

Current best public record: `submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv`.

Main Insight

A single local validation score does not fully explain Kaggle generalization.

- Weather-only and long-term weather models looked strong locally but were weaker on the public leaderboard.
- 91-day-gapped score history still provides important region-specific context.
- CatBoost's native categorical handling added useful ensemble diversity and improved public MAE.
- CatBoost 35%, 40%, and horizon-ramp probes did not beat the 30% public anchor.
- Rolling-origin MAE is useful, but should not be compared directly against older holdout MAE.

The next decision should be based on private robustness, not only public score.

Next Steps

- Compare the LGB/XGB anchor and CatBoost blends on the 5/22 static private leaderboard.
- Treat the 5/20 CatBoost and LGBM reruns as post-readout blend inputs, not new anchors.
- Use `docs/current_state.json` and `scripts/check_current_state.py` to keep report, slides, and experiment records aligned.
- Finalize the IEEE report and keep code, experiments, and submission lineage consistent.

Current status: repo-recorded best legal public MAE = **0.8124**.